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The Role of Renewable Energy Communities in the Energy Transition: Opportunities and Barriers

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Abstract: Renewable energy communities (RECs) have emerged as one of the most promising instruments for advancing the social, economic, and environmental dimensions of the energy transition. Based on the principles of open participation, non-profit governance, and local value creation, RECs hold significant potential to alleviate energy poverty, revitalize inland areas, foster local employment, enhance the social acceptance of renewable energy infrastructure, and support sustainable mobility. Despite a well-defined regulatory framework, the deployment of RECs in Italy remains limited in both scale and scope. Current configurations are typically characterized by fewer than two generation units, a median installed capacity of approximately 12 kW, and around ten members. The paper aims to demonstrate that the gap between the theoretical potential of REC and their current implementation reflects a structural issue, which has led to the proliferation of models driven primarily by incentive schemes rather than by strategic, community-centered planning. Through a systematic review of the multidimensional benefits associated with RECs and an analysis of the structural barriers hindering their development, the study identifies the aggregator as a key missing organizational enabler. This actor can bridge citizens, institutions, industry, and the public sector, thereby translating regulatory intent into sustainable and scalable community energy initiatives.

Keywords: Renewable energy community; Aggregators; Business model; Energy poverty; Inland area; Sustainable mobility; Short chain

1 Introduction

The Intergovernmental Panel on Climate Change (IPCC) has highlighted the central role of greenhouse gases (GHG) and anthropogenic causes in driving climate change [1] . Despite this well-documented scientific consensus, global carbon dioxide (CO_2) emissions continue to rise, reaching a record value of 37.4 billion tons in 2023 [2]. In parallel, recent years have also exposed a structural fragility in energy systems that extends beyond the environmental concerns. The energy crisis that affected Europe has demonstrated that even advanced economies remain vulnerable to supply disruptions, price volatility, and insufficient diversification of energy sources [3]. In response, the European Union (EU) has developed an ambitious regulatory framework aimed at enabling a deep transformation of the energy sector, while also addressing social cohesion and solidarity. Within this context, Renewable energy communities (RECs) have emerged as a key instrument for tackling the social, economic, and environmental dimensions of the just transition [4]. Grounded in the principle of open and voluntary participation, RECs actively involve citizens, including vulnerable households, and are designed to generate environmental, economic, and social benefits for their members and local communities, rather than financial profits [5]. At the same time, the REC regulatory framework defines guiding principles rather than a fixed organizational structure, thereby enabling a wide range of application scenarios. These include electric vehicle integration and shared mobility schemes, the provision of ancillary services to the power grid, the optimization of end-user energy consumption, and the development of synergies between industrial sites and peri-urban contexts [3, 4]. While this flexibility represents a key strength, allowing context-specific solutions, it also introduces complexity, leaving room for interpretation and potentially leading to partial or suboptimal implementations.

Despite their significant potential to reshape local energy systems, the deployment of RECs in Italy remains limited in scale and largely driven by incentive mechanisms rather than strategic, community-oriented planning. This has resulted in fragmented configurations that do not fully exploit their collective benefits. From this arises concern about whether RECs are living up to their transformative potential or whether the current model systematically falls short of the social and solidarity objectives embedded in their definition. As of 31 December 2025, approximately 900 RECs have been registered on the Gestore dei Servizi Energetici (GSE, Italian regulatory entity) platform; however, most configurations include only 1–2 renewable energy sources (RES) based plants, primarily photovoltaic (PV) systems, with a median installed production capacity of 12 kWp, and an average of around 10 members [6] These figures suggest that although RECs are gradually expanding, they have yet to become deeply embedded within the broader socio-economic fabric. A key factor underlying this limitation is the absence of a structured intermediary actor, such as aggregators, capable of coordinating stakeholders, bridging gaps between citizens, industry, public authorities, and local institutions, and translating the regulatory framework into effective community-driven initiatives.

1.1 Research Aim

Against this backdrop, the paper pursues a twofold aim. First, it aims to provide a systematic overview of the multidimensional benefits that RECs can deliver across environmental, economic, and social dimensions, with particular attention to their potential as instruments of territorial revitalization in inland areas. As the Italian legislative framework makes clear, these territories were not an afterthought in the design of REC policy: the exclusion of large industrial actors from eligible membership, the non-profit nature of the scheme, and the provision of non-repayable capital grants specifically targeting municipalities below 5000 inhabitants, subsequently extended to those below 50000, all point to a deliberate attempt to make RECs a genuinely place-based tool for areas at risk of depopulation and economic marginalization [7, 8]. These territories represent both the most fertile ground for REC development and the most vulnerable to the fragilities of an incentive-driven approach, as they are rich in renewable resources but often lack the technical capacity to manage complex energy configurations.

Second, building on this analysis, the paper argues that realizing the full potential of RECs, particularly in contexts where technical expertise and institutional capacity are limited, requires moving beyond the current incentive-dependent model towards a more structured form of governance. In this regard, the role of the aggregator emerges as central: as an intermediary capable of bridging citizens, local institutions, industry, and the public sector, the aggregator represents the missing link between regulatory intent and effective community-driven implementation.

The remaining part of the paper is structured as follows: in the Section 2, the social and solidarity principles that had shaped the national and European definition of RECs are illustrated, and then a detailed description of the benefits achievable by RECs configuration is carried out, starting from the social acceptability of renewable energy based plant, moving to energy poverty mitigation, and finally leading to the revitalization of inland areas; Section 3 focused on the limits of the incentive-based policies that lays behind the RECs implementation, and then provides an overview on the crucial role that aggregator potentially have in the REC exploitation; Section 4 illustrates the prospective evolution of RECs in the wider energy sharing district configuration; and finally Section 5 reports the main conclusion of the study.

2 Benefits and Contributions of Renewable Energy Communities

The concept of REC, along with the broader notion of Energy Community, was formally introduced into European law through the Clean Energy for All Europeans Package (CEP) 2019, specifically via the Renewable Energy Directive 2018/2001 (RED II) and the Internal Electricity Market Directive 2019/944 (IEMD) [5, 9, 10]. Article 22 of RED II defines REC as a legal entity grounded in open and voluntary participation, autonomous in its governance, whose shareholders or members may be natural persons, small and medium-sized enterprises (SMEs), or local authorities, including municipalities. Crucially, the primary purpose of REC is to deliver environmental, economic, and social benefits to its members or the local areas in which it operates, not to generate financial profits. Within this framework, members are entitled to produce, consume, store, and sell renewable energy, and to share energy generated by community-owned plants among themselves [5].

In Italy, the transposition of RED II through Legislative Law 199/2021 [8] established one of the most comprehensive national frameworks for REC in Europe [11], enabling a wide range of organizational configurations, from community-scale residential REC to those promoted by industrial bodies or hybrid models combining both. The virtual sharing model, which defines how renewable energy is shared among REC members, is regulated by Testo Integrato Autoconsumo Diffuso (TIAD), published by Autorità di Regolazione per Energia Reti e Ambienti (ARERA, Italian Energy Regulatory Authority) [12], while the economic support scheme is set out in the Ministerial decree 414/2023 [7]. The operational rules for accreditation on the GSE institutional platform complete the regulatory architecture [13].

Beyond their technical and regulatory dimensions, RECs embody a broader paradigm shift: from a centralized energy system reliant on fossil fuels to one characterized by distributed polygeneration based on RES and high-

efficiency technologies. Equally significant is the governance model they introduce, a new administrative paradigm based on a democratic, bottom-up, and participatory approach, in which citizens are no longer passive recipients of an externally driven energy transition, but active agents in shaping it [14].

2.1 The Social and Solidarity Vocation of Renewable Energy Communities

A close reading of both the European and Italian regulatory frameworks reveals that the social and solidarity dimension of RECs is not a secondary feature but a foundational principle, embedded in the legislative architecture from the outset. Article 31/1/d of Legislative Decree No. 199/2021, and Article 22/4/f of RED II both establish that participation in a REC must be open, free, and voluntary, and explicitly extend to low-income and vulnerable households. This provision is not merely procedural: it signals a deliberate policy choice to design RECs as inclusive instruments, accessible to precisely those who are most exposed to the costs of the energy transition.

This intent is articulated even more explicitly in the recitals of RED II. Recital 67 argues that by granting collective self-consumption rights, RECs should be able to increase household energy efficiency and help address energy poverty through reductions in both consumption and resulting energy bills, and Member States should take appropriate steps to ensure that households, which might otherwise be excluded, including vulnerable consumers and tenants, are enabled to participate.

This social orientation is further formalized at the level of definition: Article 31/1/a of Legislative Decree No. 199/2021 and Article 2/16/c of RED II explicitly establish that the primary purpose of a REC is not the generation of financial profit, but the delivery of environmental, economic, or social benefits to its members and to the local areas in which it operates.

Recital 70 of RED II further highlights how the involvement of local citizens and authorities in REC projects generates significant added value: not only in terms of local acceptance of renewable technologies, but also through the mobilization of private capital, local investment, and broader civic participation in the energy transition.

This social orientation is ultimately crystallized in the operational rules published by the GSE, which translate the normative principles into binding financial constraints. In the case where the community includes business members, any financial remuneration recognized on exceeding 55% of the total shared energy, or 45% when capital grants have been accessed, may not be redistributed to them despite their significant contribution in achieving energy sharing of the whole configuration. It must instead be allocated to social purposes that benefit the local areas in which the REC operates, or to other REC members' typologies. In this sense, the regulatory framework does not merely permit a solidarity-oriented use of RECs: it requires it.

Taken together, these provisions give a coherent picture. RECs were not designed as a generic instrument of energy decarbonization: they were conceived as a place-based, people-centered model in which the pursuit of environmental goals is inseparable from commitment to social equity and territorial cohesion. It is this structural alignment between regulatory intent and social purpose that positions RECs also as a meaningful vehicle for advancing the United Nations' 2030 Agenda for Sustainable Development. The 2030 Agenda establishes the Sustainable Development Goals (SDGs) [15] as a global framework for addressing interconnected economic, social, and environmental challenges through clearly defined targets. Within this framework, RECs play a pivotal role in driving the energy transition while contributing to multiple SDGs simultaneously. In particular, they support SDG 7, "Ensure access to affordable, reliable, sustainable and modern energy for all", by lowering energy costs and fostering local development opportunities. SDG 13 "Take urgent action to combat climate change and its impacts" is advanced through local distributed renewable energy generation and reductions in GHG emissions. RECs also advance SDG 11, "Make cities and human settlements inclusive, safe, resilient and sustainable", by enabling decentralized urban energy infrastructures, while promoting economic growth and innovation. Furthermore, by facilitating the inclusive participation of vulnerable groups in energy production and sharing, RECs contribute directly to SDG 1, "End poverty in all its forms everywhere", and SDG 10, "Reduce inequality within and among countries", reflecting the explicit mandate of RED II to make the energy transition accessible to all [16].

2.2 Renewable Energy Communities' Benefits

The following sections examine the principal benefits that RECs can generate across four interconnected dimensions: social acceptability of renewable energy infrastructure, energy poverty mitigation, inland area revitalization and local employment generation, and energy sharing, including electric mobility integration. While each dimension is addressed separately for analytical clarity, these benefits are not independent variables but mutually reinforcing. Higher local employment reduces household economic vulnerability, which, in turn, mitigates energy poverty; broader social acceptance of RES installations accelerates deployment, thereby strengthening the energy and economic base of local communities. This systemic interdependence is not incidental: it reflects the place-based, multi-objective nature of the REC model itself.

2.2.1 Social acceptability of renewable energy infrastructure

As outlined in Section 2.1, one of the expected outcomes of the wider proliferation of RECs is an increase in the social acceptability of renewable energy infrastructure. Indeed, alongside technical and economic barriers, the deployment of RES infrastructure consistently encounters a form of social resistance that has become one of the most studied in energy policy research: the Not In My Backyard (NIMBY) effect. The NIMBY phenomenon is a long-standing problem that initially was associated with opposition to fossil-fuel-powered plants and has progressively extended to renewable energy infrastructure as well. As documented in the literature, it stems from a paradox: people broadly support the installation of renewable energy plants in principle but tend to oppose them when located in the vicinity of their homes or frequently visited places [17]. The consequences are tangible: this form of social resistance has contributed to substantial underinvestment in renewable energy infrastructure and, more broadly, in energy assets such as transmission lines and storage facilities [18].

The drivers of the NIMBY phenomenon span multiple dimensions, ranging from ideological concerns to economic, governance, and place-identity considerations. Among the most frequently cited are the perceived negative impact on local flora and fauna, the aesthetic degradation of the landscape, and the consequent reduction in the area's appeal to the tourism sector [17]. Another critical aspect, however, is governance: when citizens do not feel genuinely involved in the energy transition but instead perceive themselves as its passive recipients, with decision-making power that is merely superficial, opposition tends to harden, leading to the failure of the project [19]. In this sense, what drives resistance is not the technology itself, but the governance model through which it is introduced.

Evidence from a large-scale choice experiment across Austria, Germany, Italy, and Switzerland confirms that energy community configurations significantly increase local acceptance compared to conventional top-down deployment models [20]. Consistently, analysis of survey data in [17] revealed that public acceptance of solar panel installations increases when residents are actively involved in the process, when the environmental benefits and the contribution to climate change mitigation are clearly communicated, and when tangible financial incentives are offered. Furthermore, plant scale also matters: citizens tend to be more supportive of small-scale installations than larger ones. RECs address all these factors simultaneously. Their bottom-up governance model places citizens at the center of decision making process rather than at the receiving end of it; their regulatory framework caps individual plant capacity at 1 MW, aligning with the scale preferences documented in the literature and, through the national economic support scheme, they generate both direct and indirect economic benefits for members; in some cases, as discussed further below, they also contribute to local attractiveness through the integration of electric vehicles charging infrastructure. The accreditation data of REC on the GSE platform confirms these considerations. As of 31 December 2025, a total of 1429 power plant installations have been set up, with a combined capacity of approximately 95 MW. The dominant technology is PV, though exceptions exist, such as the wind-based REC of Castiglione (Central Italy) [21].

2.2.2 Energy poverty mitigation

Energy poverty represents one of the most pressing social challenges associated with the current energy transition. Several political and economic shocks in recent years, from the COVID-19 pandemic to the war in Ukraine and the ensuing energy crisis, have unveiled a structural fragility that extends well beyond the developing world, affecting households across highly developed nations. Despite its growing prominence in worldwide policy debates, energy poverty remains a concept with unclear contours: no universally accepted definition exists, and this uncertainty has long hampered the development of a shared methodological framework for its analysis and monitoring [3]. The complexity of the phenomenon reflects the multidimensional nature of its determinants. Household income, local climatic conditions, energy prices, and broader socioeconomic factors, including health status, literacy, and degree of social integration, all interact in shaping a household's energy vulnerability [22]. A significant step towards conceptual harmonization was taken with the recast of the Energy Efficiency Directive (EED 2023/1791) [23], which introduced the first formal EU energy poverty definition: energy poverty is linked to the inability to access essential energy services, encompassing adequate heating, cooling, hot water, lighting, and the power needed to run basic appliances, resulting from a combination of no affordability, low income, high energy expenditure and mediocre building energy efficiency. Building on this regulatory definition, the Energy Poverty Advisory Hub (EPAH) [24] has proposed a diagnostic framework comprising 28 indicators grouped into primary and secondary categories. Primary indicators capture direct manifestations of energy poverty, such as the inability to maintain adequate indoor thermal comfort or arrears in energy bill payments; other primary indicators are expenditure-based thresholds: a household may be experiencing energy poverty if its energy expenditure falls below half the national median (M/2) or exceeds twice that value (2M) [25]. Within this backdrop, the Italian official authority, Osservatorio Italiano sulla Povertà Energetica (OIPE) [26] has published reports showing that the percentage of households experiencing energy poverty in 2023 stands at 9%, the highest rate ever recorded [27].

Within this framework, RECs have demonstrated concrete potential to address this challenge. A case study focused on a small Energy Community in Southern Italy, involving three households sharing a PV roof system, shows measurable improvements across standard energy poverty indicators, with both the 10% threshold and the Indicatore

della Situazione Economica Equivalente (ISEE, equivalent economic situation indicator), based index falling well below the critical levels, following participation in the community [28]. Similar findings emerge from a Spanish context: the Energy Community established in Getafe, on the outskirts of Madrid, enables tangible reductions in electricity expenditure for households experiencing energy poverty [29].

These results find a direct echo in the Italian experience of so-called solidarity RECs, configurations explicitly designed to prioritize the inclusion of vulnerable households. A prominent example is the community established in San Giovanni a Teduccio (Naples), Comunità Energetica e Solidale di Napoli Est, recognized as Italy's first energy and solidarity community [30]. Promoted by Legambiente Campania in partnership with the non-profit organization Famiglia di Maria, which made its rooftop available for a 53 kW PV system, the community shared renewable energy with 40 families, demonstrating how the REC model can be operationalized as a direct instrument of social solidarity.

The potential of RECs as solidarity instruments has also been recognized by institutional actors beyond the public sector. The Italian Episcopal Conference (Conferenza Episcopale Italiana, CEI), in collaboration with the GSE, has published a dedicated guide to support churches and religious organizations in establishing RECs, framing the initiative within the broader ecological and social teaching of Pope Francis [31]. Building on these considerations, the Diocesan Foundation REC-La Pace, in Benevento (Southern Italy), was formally established in May 2025, with the explicit mission of generating environmental and social benefits for the diocesan territory, with particular attention to families experiencing economic hardship [32].

2.2.3 Local resource valorization, employment generation, and inland area development

The depopulation of inland areas represents one of the most pressing territorial challenges facing Italy. The ongoing exodus from southern regions and from peripheral municipalities, compounded by a persistent decline in birth rates, points to the progressive demographic and economic marginalization of territories that nonetheless account for 60% of the country's total land area, 52% of its municipalities, and 22% of its population [33]. These areas are frequently characterized by a profound paradox: on one hand, they are rich in RES, solar radiation, wind, biomass from forest management, that remain largely unexploited due to low local consumption levels; on the other, they suffer from a chronic shortage of technical expertise and institutional capacity, which limits their ability to design and manage complex energy systems. RECs can help address this contradiction by activating short-supply-chain mechanisms that transform locally available resources into economic and energy value for the community. A particularly significant example in this context is the valorization of woody biomass from forest maintenance activities. The case study presented in [34] analyses the feasibility of a REC based on a cogeneration system fueled by residues from forest management, sawmill processing, and walnut shell waste, demonstrating how materials that would otherwise be treated as waste can be converted into a stable energy source for local communities while simultaneously reducing maintenance costs and fire risk in forested areas.

Beyond woody biomass, inland areas, especially in Southern Italy, offer substantial potential for biogas-based RECs drawing on agro-zootechnical waste streams. The anaerobic digestion of livestock manure and dairy processing residues, such as cheese whey, represents a well-established pathway for distributed energy generation in rural contexts. In recent years, the deployment of biogas plants has proven particularly effective in rural areas, generating benefits not only for individual farms but for entire communities, contributing simultaneously to reductions in energy poverty and to the production of both energy and organic fertilizer [35]. From an agronomic standpoint, the digestate produced by anaerobic digestion can replace chemical fertilizers, close the nutrient cycle, and generate additional economic value for participating farms. Research focused on southern Italy has also explored the co-digestion of olive oil by-products and cheese whey as a viable feedstock option [36], pointing to the potential for RECs that integrate multiple local agro-industrial waste streams within an energy configuration.

The development of these local supply chains carries significant employment implications. The GSE's Energy and Climate Report [37] documents that 2022 saw measurable growth in both temporary and permanent employment generated by the renewable energy sector. Temporary positions, both direct and indirect, are primarily linked to the construction and installation of new RES-based plants; permanent roles stem from the operation, maintenance, and management of existing infrastructure. PV technology, which dominates current REC configurations, has been a particularly significant contributor to both categories. Beyond direct employment, the activation of local biomass and biogas supply chains generates indirect occupational effects across the agricultural, forestry, and maintenance sectors, creating a multiplier effect that extends well beyond the energy system itself.

In this sense, RECs in inland areas are not merely energy configurations: they are instruments of territorial development, capable of converting underutilized local resources into stable economic activity and of retaining populations at risk of marginalization.

2.2.4 Electric mobility and sustainable transport

The concept of ownership has been undergoing a profound transformation. The sharing economy has taken hold across multiple sectors, from housing to transport, with a growing number of people choosing to access services rather than own physical assets. This trend is also relevant in the transport sector, which remains one of the largest

contributors to CO_2 emissions, and is widely regarded as a hard-to-abate sector due to the structural complexity of its decarbonization. Against this backdrop, two complementary strategies have gained significant relevance: promoting car-sharing and car-pooling schemes, and electrifying transport through the widespread adoption of battery electric vehicles (BEVs).

The environmental benefits achievable using BEVs are widely debated in the scientific literature: while the elimination of direct combustion processes yields a reduction in local emissions, the overall environmental impact of electric vehicles depends critically on the carbon intensity of the electricity grid used for charging [38]. A study that focused on specific regions of southern Italy, such as Sardinia and Sicily, since the carbon intensity of the local power grid, revealed that the adoption of BEVs is not sustainable from an environmental standpoint [39]. However, by integrating BEV charging infrastructure within a REC, members can ensure that their vehicles are powered by locally generated renewable electricity. It is a solution that simultaneously improves the environmental impact of electric transport and increases the amount of shared energy remunerated under the national incentive framework [40]. When combined with car-sharing or car-pooling schemes managed within the REC, the benefits multiply further. Car-sharing fleets that transition to electric vehicles can considerably reduce both air and noise pollution in urban areas, two factors that significantly impact public health and quality of life [41]. Beyond these, shared mobility schemes help reduce traffic volumes and improve transport access for the most vulnerable users, including elderly residents and those without driving licenses, who might otherwise be excluded from mobility services in areas with limited public transport provision.

A further, often overlooked benefit of integrating electric vehicles charging infrastructure within a REC is its potential contribution to local attractiveness. As the number of BEV users continues to grow, the availability of charging points is increasingly becoming a factor in travel planning decisions. Communities that install accessible charging infrastructure may find themselves included in the itineraries of BEV-owning tourists who would otherwise bypass them. In this sense, investing in REC-integrated charging infrastructure is not only an energy decision: it is also a territorial development and tourism strategy.

3 From Potential to Practice: Structural Constraints and the Role of Aggregators

The limited scale of RECs deployment in Italy can not be attributed to a single bottleneck. Rather, it reflects a combination of structural, technical, administrative, and cultural barriers that interact and reinforce one another, making the transition from regulatory intent to effective implementation significantly more complex than the framework design might suggest.

3.1 The Limits of Incentive-Based Policies

As outlined in Section 1, despite a well-structured regulatory framework, the average REC configuration in Italy consists of one to two PV plants, with a median installed capacity of approximately 12 kW, and around ten members [6]. These figures reflect a model that is currently stagnating. Understanding the underlying causes requires moving beyond technical and administrative barriers to examine the logic shaping how RECs are conceived and implemented in practice. Across the Italian energy landscape, from SMEs to local authorities and civil society organizations, the dominant approach to the energy transition remains largely reactive. Rather than emerging from a structured assessment of local needs, consumption patterns, and long-term territorial strategies, energy initiatives are often triggered by the availability of specific incentive schemes. In this context, incentives no longer act as enabling tools but increasingly become the primary objective.

This dynamic is understandable given the constraints faced by most REC promoters, including limited financial resources, insufficient technical expertise, and a lack of time to engage with the complexity of the energy system in a structured way. As a result, the transition is driven by short-term opportunities rather than strategic planning, leading to fragmented configurations: initiatives that are individually viable yet disconnected from the broader social, economic, and territorial context, and that fail to capture the multidimensional benefits envisaged by the regulatory framework.

This issue is further reinforced by a cultural dimension. The prevailing narrative surrounding RECs tends to frame renewable energy sharing as the final objective, whereas energy production should instead be considered a starting point. The principles of Energy Efficiency First and distributed generation require that production be accompanied by a corresponding transformation in consumption patterns. Without this shift, RECs risk being reduced to mere instruments for economic optimization rather than functioning as catalysts for social and territorial development. As long as incentives remain the endpoint rather than the starting point, such a transformation is unlikely to materialize.

3.2 Challenges and Structural Constraints

At the administrative level, the authorization procedures for the installation of new RES-based plants remain a significant barrier, particularly in inland and rural areas where institutional capacity is more limited and processing times are longer, thereby adding further complexity to project implementation.

Beyond the procedural aspects, a more fundamental challenge concerns the social prerequisites for a functioning REC. A community energy configuration requires, as a precondition, a heterogeneous group of actors who share not only physical proximity but a willingness to align strategies, responsibilities, and expectations around common objectives. This social network cannot be assumed; it must be actively built and maintained. Where it is absent or fragile, even well-designed RECs struggle to take root and persist over time. In this context, disengagement and low participation should not be considered peripheral risks but rather core structural vulnerabilities of the model.

A third dimension of constraint relates to the information environment. The complexity of the incentive schemes, the evolving regulatory landscape, and the proliferation of technical and contractual arrangements have created conditions in which misinformation and misaligned expectations are widespread. As a result, RECs are often framed and perceived primarily as mechanisms for accessing public subsidies, rather than as instruments enabling a broader transformation in the local production, consumption, and governance of energy. Addressing this misperception is therefore a prerequisite for fostering the long-term engagement required for successful and resilient REC development.

3.3 The Role of Aggregators in Local Energy Systems

The challenges outlined in the preceding sections, reactive incentive-driven planning, administrative complexity, weak social network, and the cultural gap between energy production and conscious consumption, do not require pure technical solutions, but rather an organizational response. What is currently missing in the REC landscape is not more regulation or further incentives, but a structuring actor capable of translating the model's transformative potential into projects that are sustainable, scalable, and durable over time. This role can be fulfilled by the aggregator.

In an evolving regulatory environment and an increasingly dynamic technological context, the diverse actors involved in a REC, households, businesses, public authorities, and third-sector organizations, require a coordinating entity capable of orchestrating competencies, aligning expectations, and distributing responsibilities in a coherent and transparent manner. Industry stakeholders and sectorial associations are particularly well-positioned to take on this function. Their ability to represent local territories, productive systems, and economic communities makes them especially suited to supporting initiatives that require an inclusive, adaptive governance structure capable of evolving alongside the community. Among the emerging configurations, business-led RECs exhibit particularly promising features. In these configurations, one or more enterprises act as initiators and anchor investors, contributing capital, technologies, services, and managerial expertise. This creates a stable initial core that supports project viability during early stages. Once this foundation is established, the REC can progressively expand to include the surrounding territory: households, public administrations, social organizations, and third-sector entities can join, share benefits, and actively participate in a collaborative framework rather than remaining passive beneficiaries. It is precisely during this phase of expansion that the aggregator's role becomes critical. Aggregators provide the governance infrastructure necessary to support scaling processes, including replicable models, standardized tools, clear contractual arrangements, and shared procedures that reduce technical and administrative barriers for new participants. At the same time, they provide operational support to non-specialist actors, enabling communities to understand opportunities, responsibilities, and regulatory requirements without compromising their autonomy or the initiative's bottom-up nature. Equally important is the aggregator's role as an interface with public administration. By maintaining a continuous dialogue with local authorities, aggregators facilitate the integration of RECs into local energy planning processes, streamline access to incentive mechanisms, and support navigation of authorization procedures that represent one of the most persistent barriers to REC development. Their territorial reach also allows them to move beyond the pilot experiences: by connecting multiple initiatives within shared organizational and procedural frameworks, aggregators enhance replicability and scalability, transforming RECs into systemic components of local energy governance.

4 Future Perspectives

The expansion of RECs holds significant potential to reshape the urban energy landscape. To fully harness this opportunity, challenges related to research, policy, and practical implementation must be addressed, as they directly influence future developments. Urban areas are major consumers of energy and contributors to GHG emissions [42]. The building sector, alongside transportation, accounts for roughly 75% of global urban CO_2 emissions [43], highlighting the urgency of enhancing urban sustainability to achieve carbon neutrality by 2050 [9]. In this context, European policy frameworks increasingly promote district and community-based approaches aimed at achieving both "zero-energy" and "positive energy" targets.

From an urban planning and energy system design perspective, RECs are indeed expected to evolve into more complex and integrated configurations, such as Positive Energy Districts (PEDs), contributing to the EU vision of climate-neutral cities [16]. This transition requires advancements in energy management and control, particularly in integrating multi-vector energy systems and smart district infrastructure. PEDs can be regarded as a natural evolution of RECs, characterized by more comprehensive governance and operational models, and by a shift from geographically limited setups to flexible, scalable spatial arrangements.

Commonly, RECs do not adhere to a standardized size, thereby their scale could be shaped by local regulations and applications. For example, Spain and France impose maximum distances between members and generation facilities [44], while in Italy [7] and Austria [44] energy sharing is typically confined to users within the same electrical substation. Consequently, urban context assessment is crucial for the implementation of these configurations. In smaller municipalities, RECs may engage the entire population while respecting regulatory limits, whereas in larger cities they may be confined to neighborhood-scale initiatives, with local characteristics strongly influencing project design.

PEDs extend the concept of energy communities not only in spatial terms but also from the energy vectors, integrating electricity, thermal, and mobility demands within a unified system [45]. This sector coupling improves overall efficiency, optimizes local resources, and enhances demand-side flexibility. By expanding both operational scope and objectives, PEDs aim not only to balance local energy consumption with renewable generation but to achieve a net-positive energy outcome. Such a shift amplifies benefits, including greater self-sufficiency, economic gains, reduced emissions, and improved resilience. Moreover, PEDs enable structured engagement with external stakeholders and neighboring energy systems, extending energy sharing beyond internal community boundaries. This outward-looking approach supports multi-actor collaboration and development of integrated energy ecosystems, aligning local initiatives with broader decarbonization goals. In this sense, PEDs mark a critical step toward systemic and cooperative energy transitions capable of delivering climate neutrality at the district scale and beyond.

The progression toward PEDs also involves expanding energy coverage beyond electricity to include heating and cooling needs [45]. Incorporating shared thermal energy is a key driver for future community development, requiring enabling infrastructures such as district heating networks (DHNs) for physical interconnection and efficient resource use. However, this evolution also introduces new constraints, including reduced user flexibility and restricting the entry of new participants unless pre-established connections exist. As a result, the scale of such configurations becomes highly site-specific, making expansion or reduction challenging.

5 Conclusions

This paper provides new insights into RECs, with a particular focus on the Italian context. Based on their regulatory framework, RECs represent innovative schemes with the potential to meaningfully reshape local energy systems; however, they have so far fallen short of concrete results. The analysis developed in the preceding section yields a set of conclusions that are both diagnostic and prospective.

From a regulatory perspective, the legislative architecture surrounding RECs, from RED II and the Italian transposition through Legislative Decree No. 199/2021 to the Configurazioni di Autoconsumo per la Condivisione dell'Energia Rinnovabile (CACER) Decree and the GSE operational rules, reflects a coherent and intentional policy design. RECs were conceived not merely as decarbonization tools but as place-based, people-centered instruments aimed at promoting social equity, territorial cohesion, and the inclusion of vulnerable groups. The solidarity vocation of the model is not a secondary feature: it is embedded in the founding definitions, the participation requirements, and the financial redistribution constraints that bind REC governance.

The potential benefits of RECs are multidimensional and mutually reinforcing. As outlined in Section 2, RECs can enhance the social acceptability of renewable energy infrastructure by shifting from top-down approaches to bottom-up co-ownership, mitigate energy poverty through targeted solidarity mechanism, stimulate the revitalization inland areas by activating local resource supply chains, from forest biomass to agro-zootechnical waste streams, and generating employment and also advance sustainable mobility by integrating electric vehicle and car-sharing schemes powered by locally generated renewable electricity. These benefits are not isolated but interconnected within the place-based logic of RECs.

However, the analysis also reveals a persistent gap between these potential and current outcomes. The data on accredited REC configurations in Italy point to a system that is growing in number but remains limited in structural depth, characterized by minimal plant configurations, limited participation, and an incentive-capture logic rather than territorial development. The analysis in Section 3 identifies the structural roots of this gap: a reactive planning culture centered on incentives, administrative and technical barriers, and weak or underdeveloped local social networks. Addressing these limitations requires an organizational, rather than purely technical or regulatory, response. In this context, the aggregator emerges as a key enabling actor within the REC ecosystem. Whether embodied by industry associations, public entities, or specialized intermediaries, aggregators can provide the structural capacity needed to build and sustain RECs: fostering stakeholder coordination, navigating administrative complexity, delivering replicable governance models, and connecting individual initiatives into scalable territorial systems. Business-led RECs, in which enterprises act as initial investors and facilitators before gradually opening to broader community participation, represent a promising pathway for implementation, especially in contexts with limited institutional capacity.

Looking ahead, the evolution of RECs toward PEDs and their contribution to the SDGs outlined in the UN 2030 Agenda suggest that the current REC configurations should be understood not as an endpoint but as a foundational

step. Realizing this trajectory, however, requires a fundamental shift in how RECs are conceived, promoted, and governed, from instruments of incentive capture to engines of territorial and social transformation.

Author Contributions

Conceptualization, F.A., C.M., I.P., and M.S.; methodology, F.A., C.M., I.P., and M.S.; formal analysis, F.A., C.M., I.P., and M.S.; investigation, F.A., C.M., I.P., and M.S.; data curation, F.A., C.M., I.P., and M.S.; writing—original draft preparation, F.A., C.M., I.P., and M.S.; writing—review and editing, F.A., C.M., I.P., and M.S.; visualization, F.A., C.M., I.P., and M.S.; supervision, F.A., C.M., I.P., and M.S. All authors have read and agreed to the published version of the manuscript.

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Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

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Nomenclature

ARERA	Autorità di Regolazione per Energia Reti e Ambiente, Italian Energy Regulatory Authority
BEV	battery electric vehicle
CACER	Configurazioni di Autoconsumo per la Condivisione dell'Energia Rinnovabile
CEI	Conferenza Episcopale Italiana, Italian episcopal conference
CO_2	carbon dioxide
DHN	district heating networks
EED	Energy Efficiency Directive 2023/1791
EPAH	Energy Poverty Advisory Hub
EU	European Union
GHG	greenhouse gases
GSE	Gestore dei Servizi Energetici, Italian regulatory entity
IEMD	Internal Electricity Market Directive 2019/944
IPCC	Intergovernmental Panel on Climate Change
ISEE	Indicatore della Situazione Economica Equivalente, equivalent economic situation indicator
NIMBY	Not In My Backyard
OIPE	Osservatorio Italiano sulla Povertà Energetica, Italian energy poverty observatory
PED	Positive Energy District
PV	photovoltaic
REC	renewable energy community
RED	Renewable Energy Directive
RES	renewable energy source
SDG	Sustainable Development Goals
SMEs	small and medium-sized enterprises
TIAD	Testo Integrato Autoconsumo Diffuso